α-$BaCu_2Se_2$ Thin-Film XRD Analysis and Bayesian Optimization

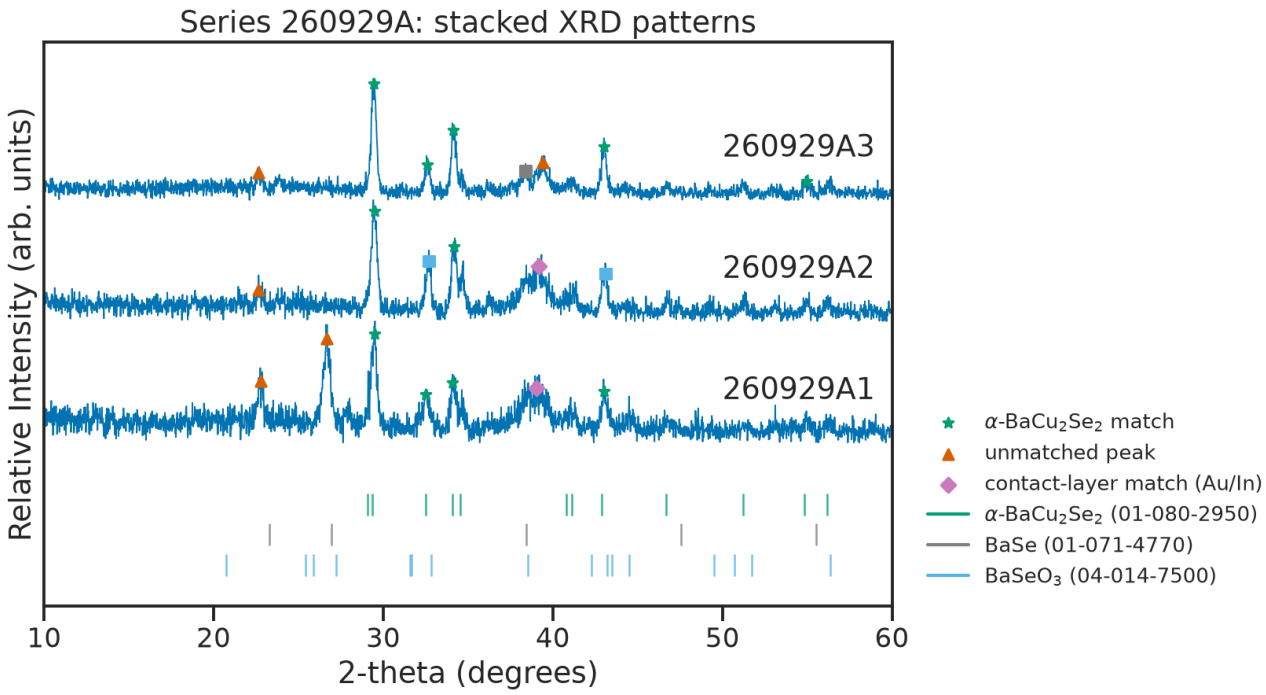
NOMAD-driven data extraction, phase matching, and GP-based synthesis planning

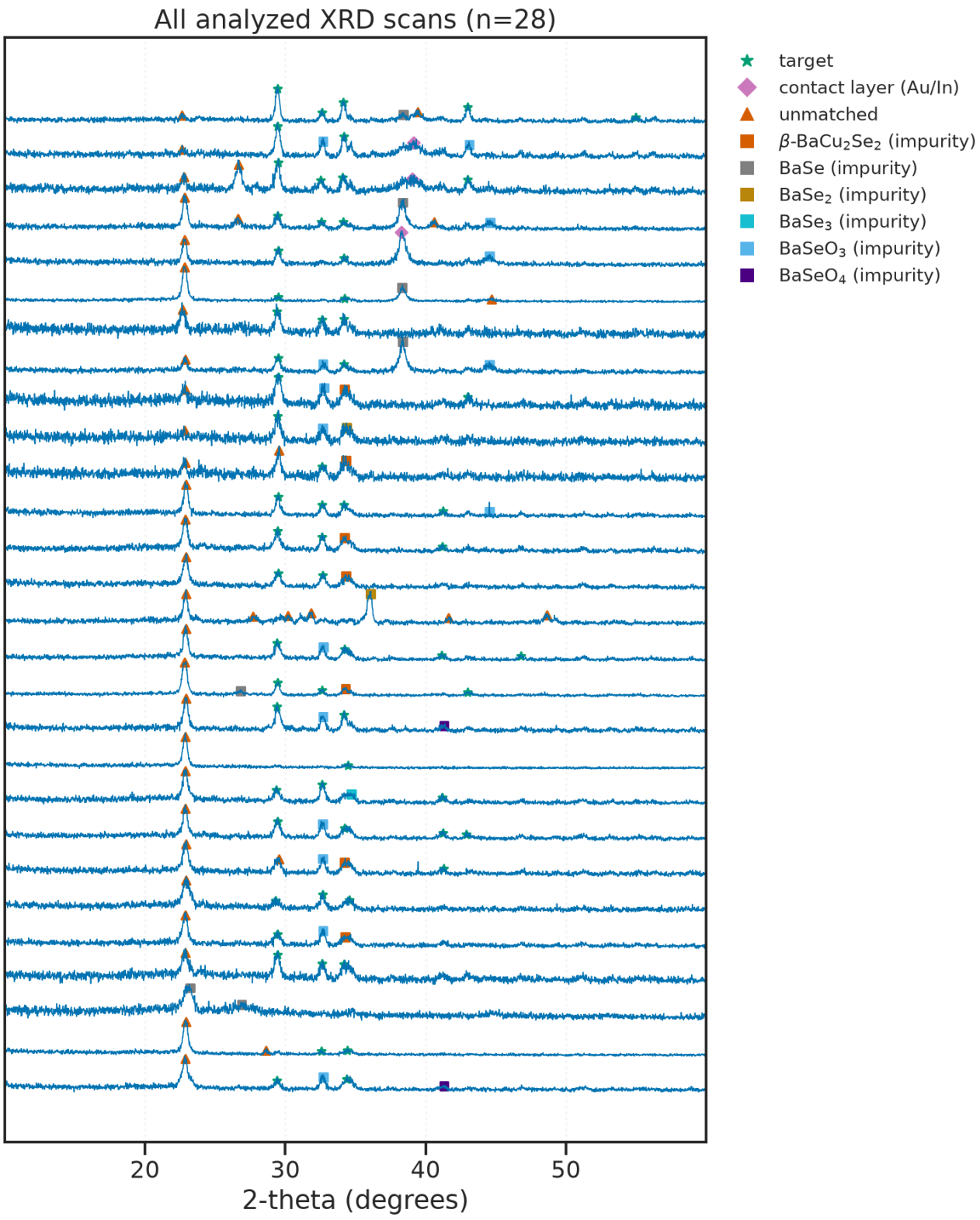
1. What was done

Synthesis records and XRD scans were pulled directly from the NOMAD Oasis REST API for the 42 α-$BaCu_2Se_2$ thin-film samples (INLThinFilmStack entries). Synthesis parameters (spin coating, soft/hard bake, selenization) and selenium-pulse settings were extracted for all samples; 35 include pulse on/off data. Each record also carries the precursor Cu/(Ba+Cu) ratio, the number of spin-coating cycles (deposited layers, excluding dynamic-dispense/seed-layer preparation steps), the Mo-coating flag, and the spin-to-selenization day gap. For the 28 films with XRD scans, diffraction patterns were matched against reference cards for the α-$BaCu_2Se_2$ target (01-080-2950) and the impurities β-$BaCu_2Se_2$ (04-008-2861), Cu_2Se (01-080-4377), CuSe (00-049-1456), BaSe (01-071-4770), $BaSe_2$ (04-004-8560), $BaSe_3$ (04-004-8559), $BaSeO_3$ (04-014-7500) and $BaSeO_4$ (04-011-7418); the α target always wins an overlap, otherwise the closest reflection decides. Au (00-004-0784) and In (00-005-0642) are classified separately as contact-layer matches with one shared plot symbol; their detected-peak intervals are excluded from phase-purity accounting and the unmatched/overlap penalty. In matching is restricted to series 260929A. Phase purity and crystallinity (1/FWHM of the strongest target match) were computed per film. A Gaussian-process Bayesian optimizer was trained on the 24 complete films to recommend the next synthesis conditions, optimizing objective = (0.7 × phase purity + 0.3 × normalized crystallinity) × (1 – 0.25 × A) over a five-input space: hard-bake temperature, selenization temperature and time, Cu/(Ba+Cu) ratio, and continuous Se pulse duty cycle. The on+off period is fixed at 3200 ms (3.2 s), so pulse frequency is fixed at 0.3125 Hz and is reported as metadata, not optimized. Intermediate duty cycles are model interpolations from three observed levels and should be treated as exploratory; recommended on-times are quantized to 1 ms increments. A long-form CSV of classified detected peaks is exported for sample-level audit. CBC is defined as $n_{Cu}/(n_{Ba}+n_{Cu})$; the candidate grid is 0.63–0.71, so a ratio of 1 (no Ba) is excluded. A is the unmatched/overlap share of detected peak-interval area, a screening proxy—not a quantitative phase fraction or a confirmed assignment of the ~23° feature.

2. XRD phase identification

Stacked XRD patterns per series with reference bars for the α target (always shown) and only those impurity phases matched in the series' patterns. Markers flag target (star), impurity (square, colored by phase), shared Au/In contact (diamond), and unmatched (triangle) peaks. Au/In reference-card bars are omitted.

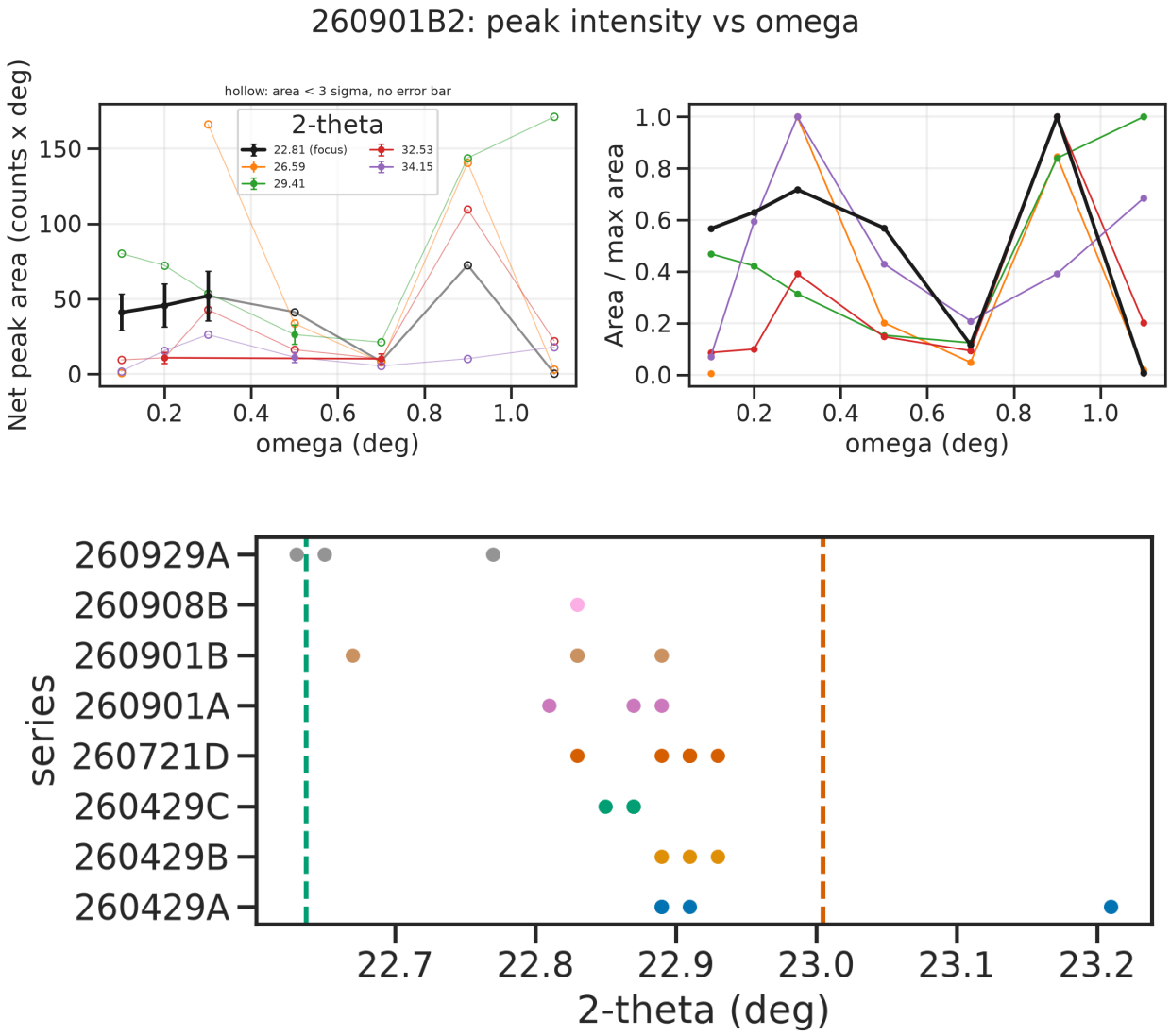

Series 260929A: stacked XRD patterns
Relative Intensity (arb. units)
260929A3
260929A2
260929A1
10
20
30
40
50
60
2-theta (degrees)
α-BaCu2Se2 match
unmatched peak
contact-layer match (Au/In)
α-BaCu2Se2 (01-080-2950)
BaSe (01-071-4770)
BaSeO3 (04-014-7500)



Omega study of sample 260901B2

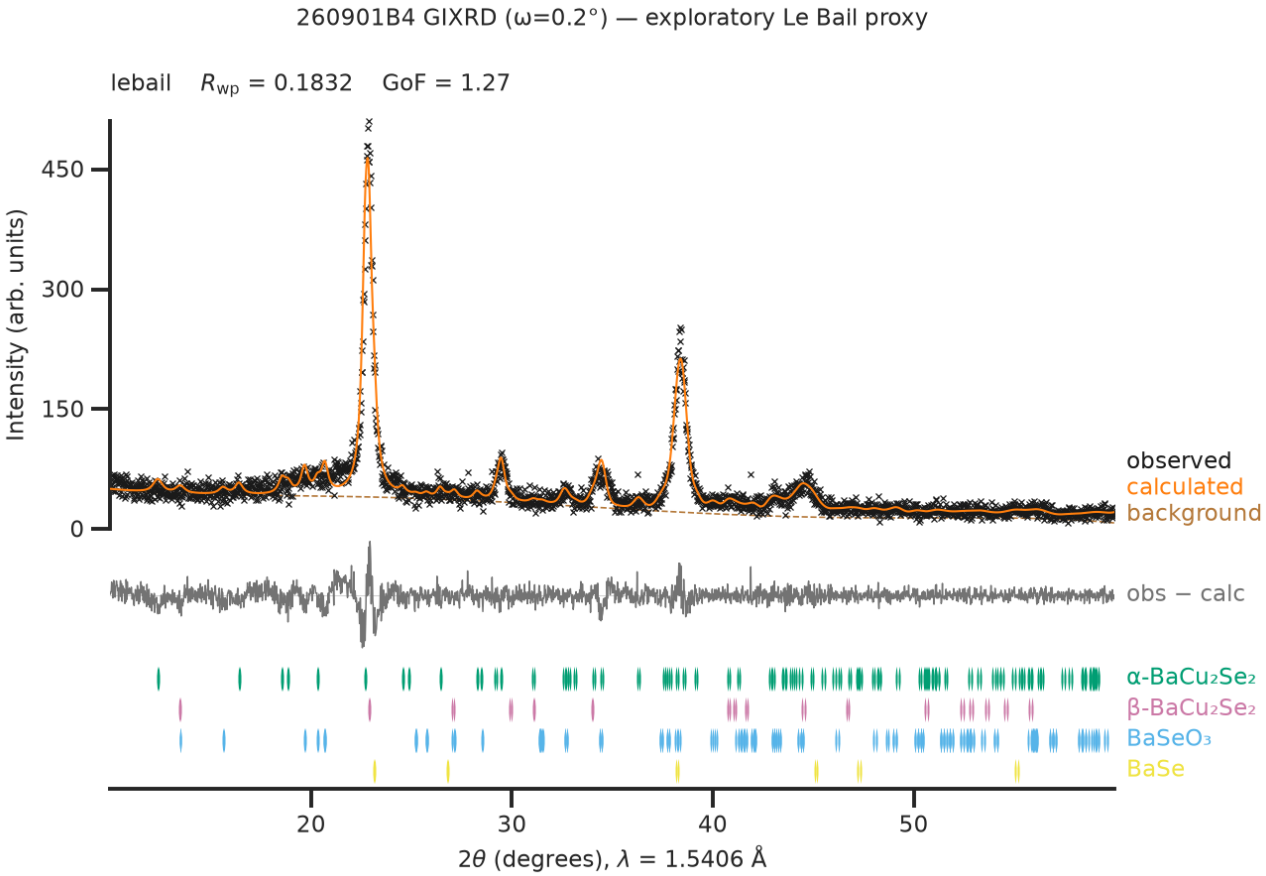
Seven grazing-incidence scans (omega 0.1–1.1°, 20–35° 2θ) were fitted peak by peak to test the origin of the ~23° feature (notebook 2.1; this sample only, not used by the optimizer). The feature sits at 22.81° 2θ with FWHM ≈ 0.4–0.5°. The 10 %-intensity matching used for purity does not assign it, but the cards list weak lines near it: α-$BaCu_2Se_2$ at 22.64° (4.6 %) and β-$BaCu_2Se_2$ at 23.01° (3.0 %), 0.17° below and 0.20° above the sample 260901B2 peak. Across all films the feature sits at 22.86–22.98° in series 260429A/B/C, 260721D and 260901A (closer to the β line) and at 22.68° in series 260929A (closer to the

α line), but proximity alone cannot establish the phase. The researcher reports that the SLG substrate has no sharp peak at 23°. Relative intensities differ from powder cards; texture is one possible explanation, but background, overlapping peaks and geometry also matter. A feature near 46.6–46.9° is a possible higher-order candidate, not an established second order: the analysis selected a maximum in a preset window without demonstrating a statistically significant peak or a common origin. Longer scans, calibration, rocking curves and pole figures can help distinguish the hypotheses.



An exploratory Le Bail profile fit (Bragg–Brentano proxy, cells fixed) was run for all 28 scans. The standard set is α- and β-$BaCu_2Se_2$ plus $BaSeO_3$ and BaSe, the impurities matched in most films (Cu_2Se was never matched). It was compared with other phase sets (table). Extra phases lower the overall R_{wp} but hardly change the 22.1–23.1° window: the four-phase standard fits it as well as nine phases. β alone fits the 23° window slightly better than α alone (lower RMSE in 18 of 28 scans) but the whole pattern much worse, and α plus β fits the window best. Le Bail intensities are free and the cells fixed, so this indicates that both polymorph lines help, not which one produces the 23° peak.

Phase set	R_{wp}	GoF	23° window RMSE	|peak pos. err.| (°)	calc/obs height
5 phases (previous CIFs)	0.190	1.29	23.0	0.074	0.76
9 phases (all cards)	0.184	1.24	19.9	0.053	0.78
α, β, Cu2Se, BaSe	0.199	1.34	19.8	0.072	0.86
α, β, BaSeO3, BaSe (standard)	0.193	1.30	20.0	0.070	0.84
α only + BaSeO3, BaSe	0.196	1.32	25.1	0.095	0.82
β only + BaSeO3, BaSe	0.238	1.61	23.0	0.066	0.82



3. Main results

Across the analyzed films the mean phase purity is 0.68 and mean crystallinity is 2.16. The best film under the penalized score is 260429B2 (score 0.88; unpenalized score 1.00; purity 1.00, crystallinity 2.92, A 0.49). The optimizer uses the five selected inputs only. Other process and substrate descriptors remain available for descriptive exploration; duty cycle is modeled continuously, with the fixed pulse frequency excluded from the optimizer because it has no observed variation.

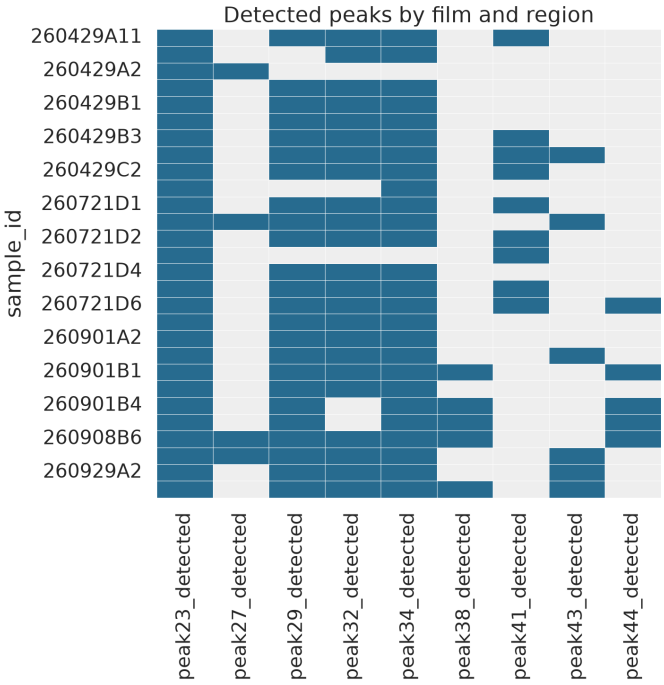
Detected XRD peaks versus synthesis

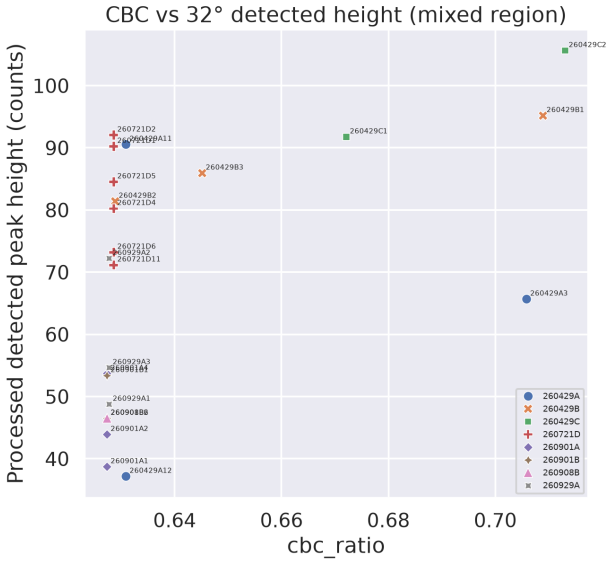
Notebook 2.2 compares detected-peak positions, FWHM, processed peak heights and detection indicators with synthesis parameters for 28 films. Peaks are grouped in selected 2θ regions, retaining the highest detected peak per region and film; undetected position, width and height are missing rather than zero. PCA and clustering use regions detected in at least 75% of films, median imputation and feature scaling. Regions may combine phases, while heights depend on measurement and processing conditions. Correlations are pairwise-complete and Benjamini–Hochberg adjusted across valid tests; shared series

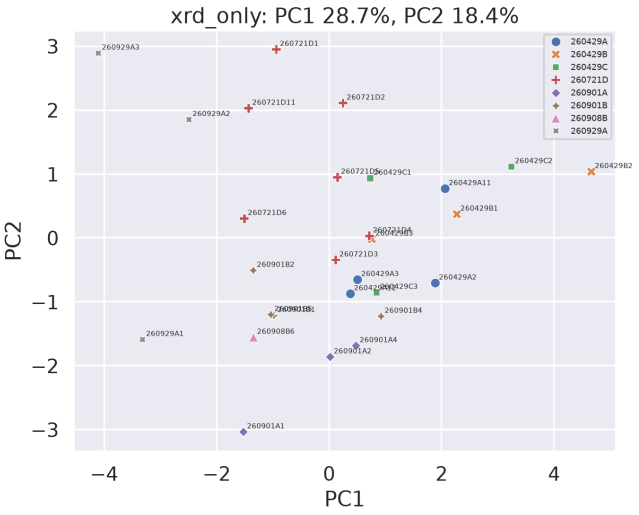
and repeated recipes violate the independent-film assumption, so p and q values are descriptive only.

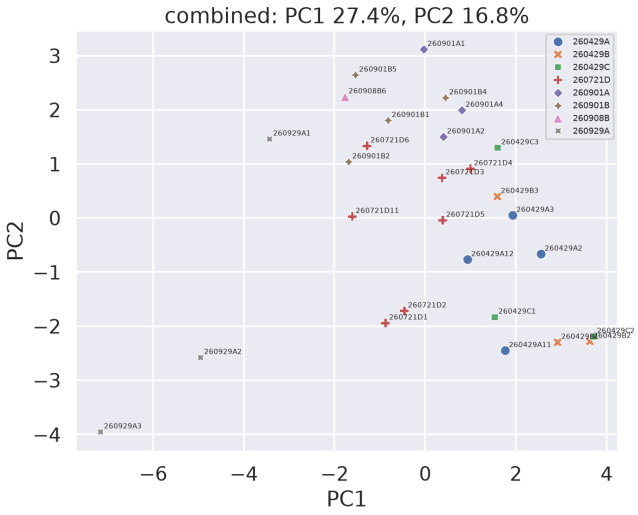
Exploratory pair	n	Spearman ρ	BH q
CBC vs 32° peak height	23	0.71	0.008
Mo coating vs 34° FWHM	26	-0.76	0.001

The 32° region contains both target and impurity labels, so its CBC association cannot identify a phase. The Mo association is strongly confounded with series; neither establishes causality. XRD-only K-means had its highest tested silhouette at k=3 (silhouette 0.15). Across 80% subsamples, adjusted Rand index has median 0.58 and 10th percentile 0.13. Low separation and feature sensitivity make the clusters exploratory rather than phase populations. Detected heights are not integrated phase intensities, and missing detections and regions outside the chosen windows limit coverage.





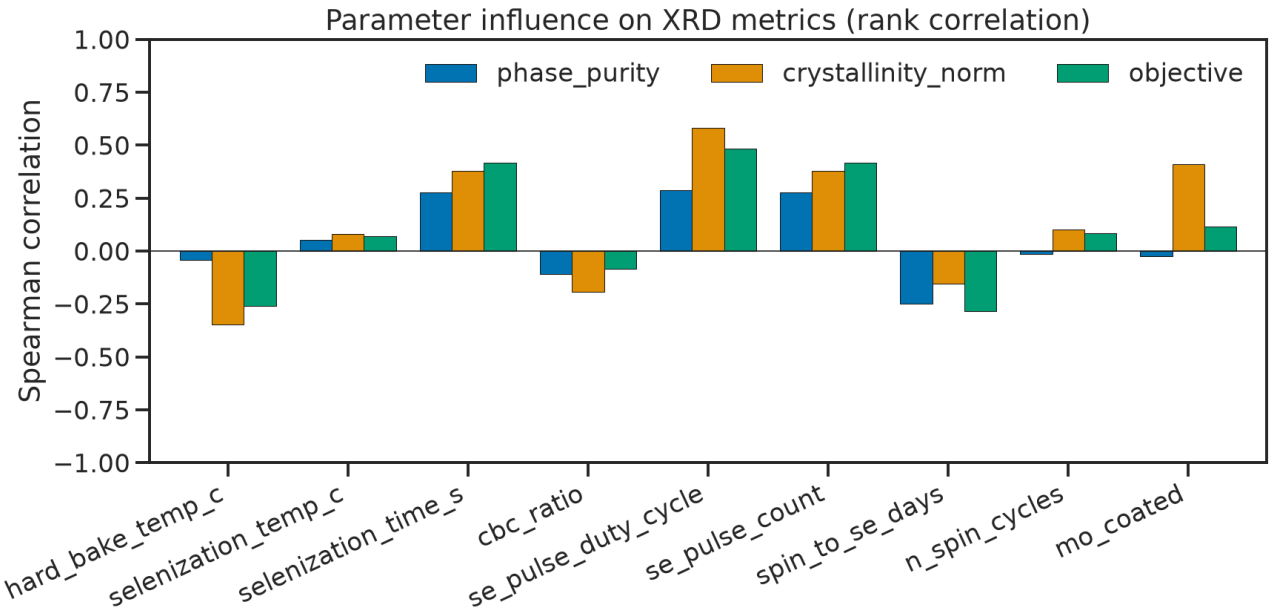


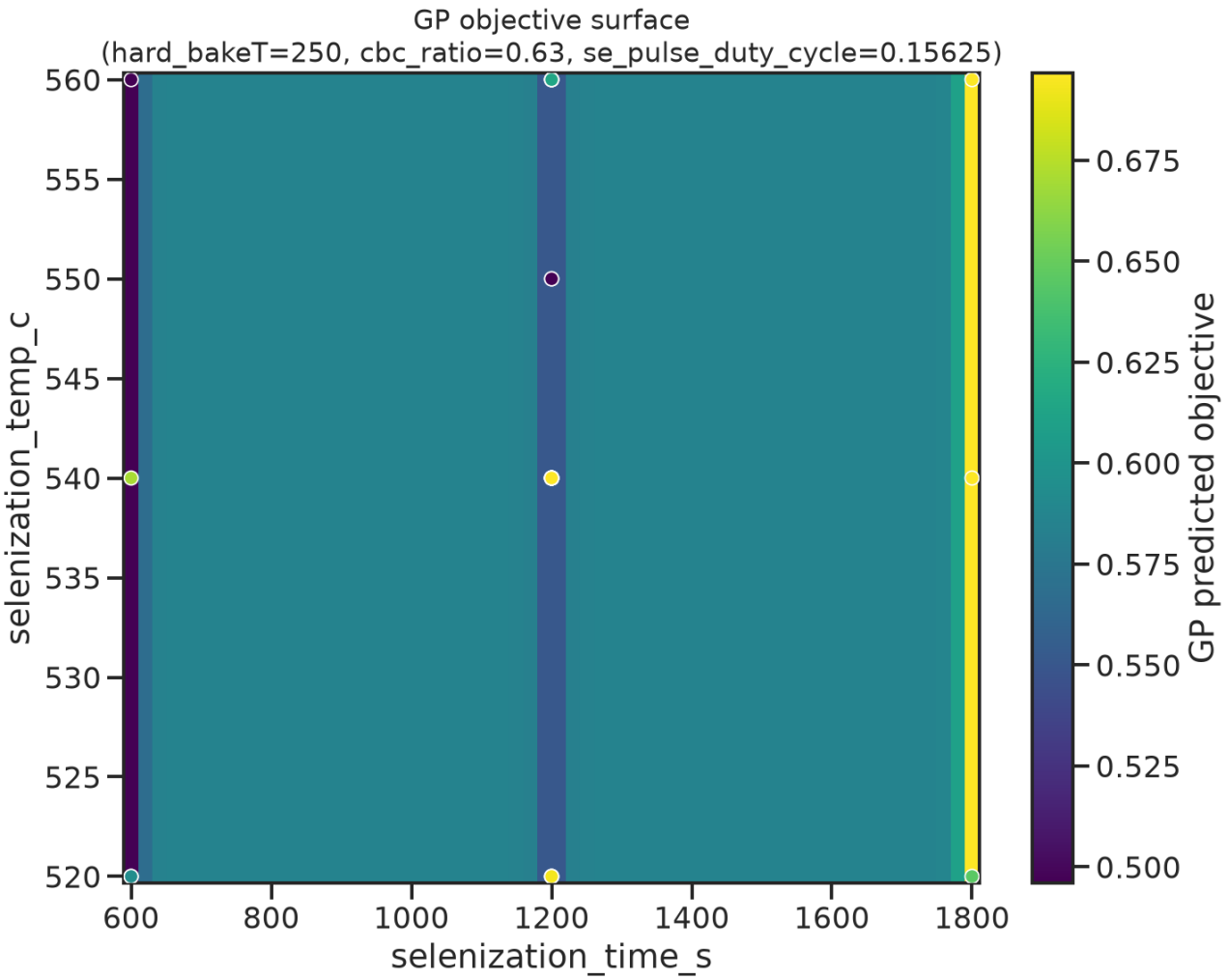


Purity–crystallinity weight sensitivity

The same films, penalty and search space were used for purity/crystallinity weights 0.5/0.5, 0.7/0.3 and 0.9/0.1. The table shows the best observed film and the first model-generated proposal in acquisition order, not a measured result. Observed score rank correlations versus 0.7/0.3 are 0.96 (0.5/0.5) and 0.93 (0.9/0.1). Recipe differences can also reflect acquisition randomness and uncertainty; seed 0 is held fixed.

Purity / crystal.	Best observed	First proposal: bake °C	Selen. °C / s	CBC	Se on / off ms
0.5 / 0.5	260429B2	225	530 / 1200	0.69	604 / 2596
0.7 / 0.3	260429B2	250	540 / 1800	0.68	500 / 2700
0.9 / 0.1	260901B2	250	560 / 1800	0.63	845 / 2355



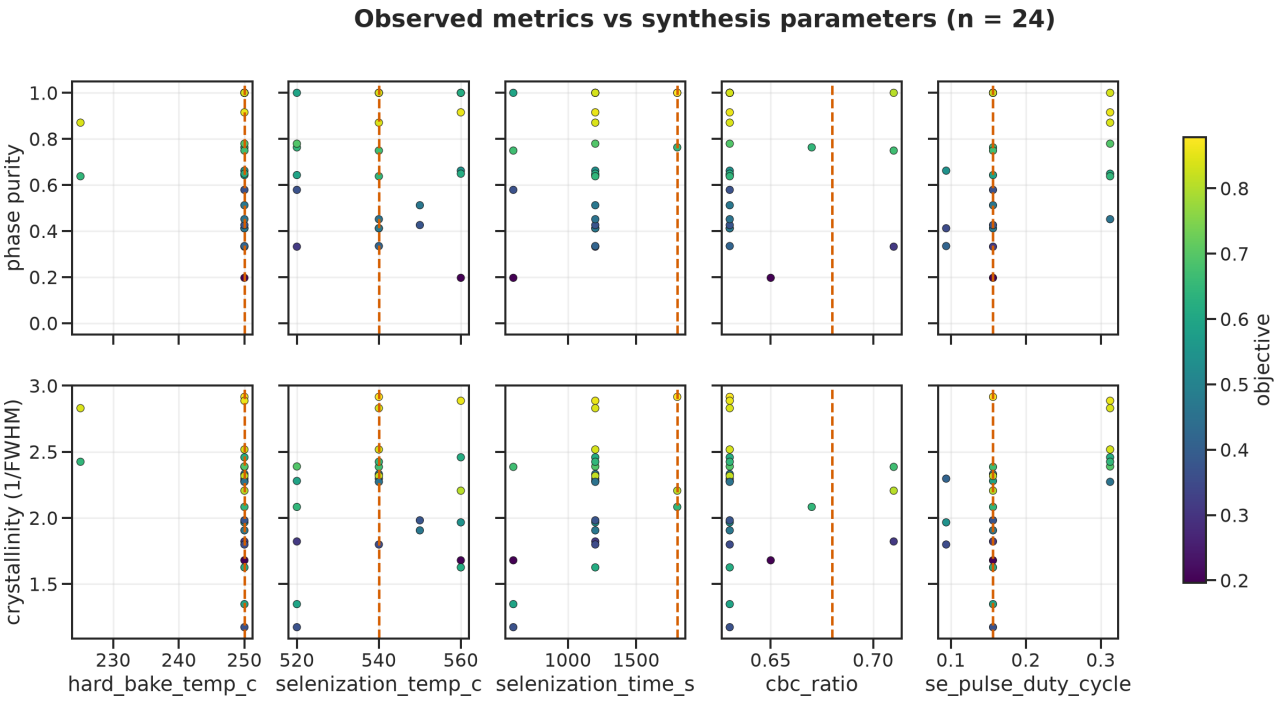


4. Suggested next experiments

Top Gaussian-process suggestions (predicted objective ± std), ranked by expected improvement:

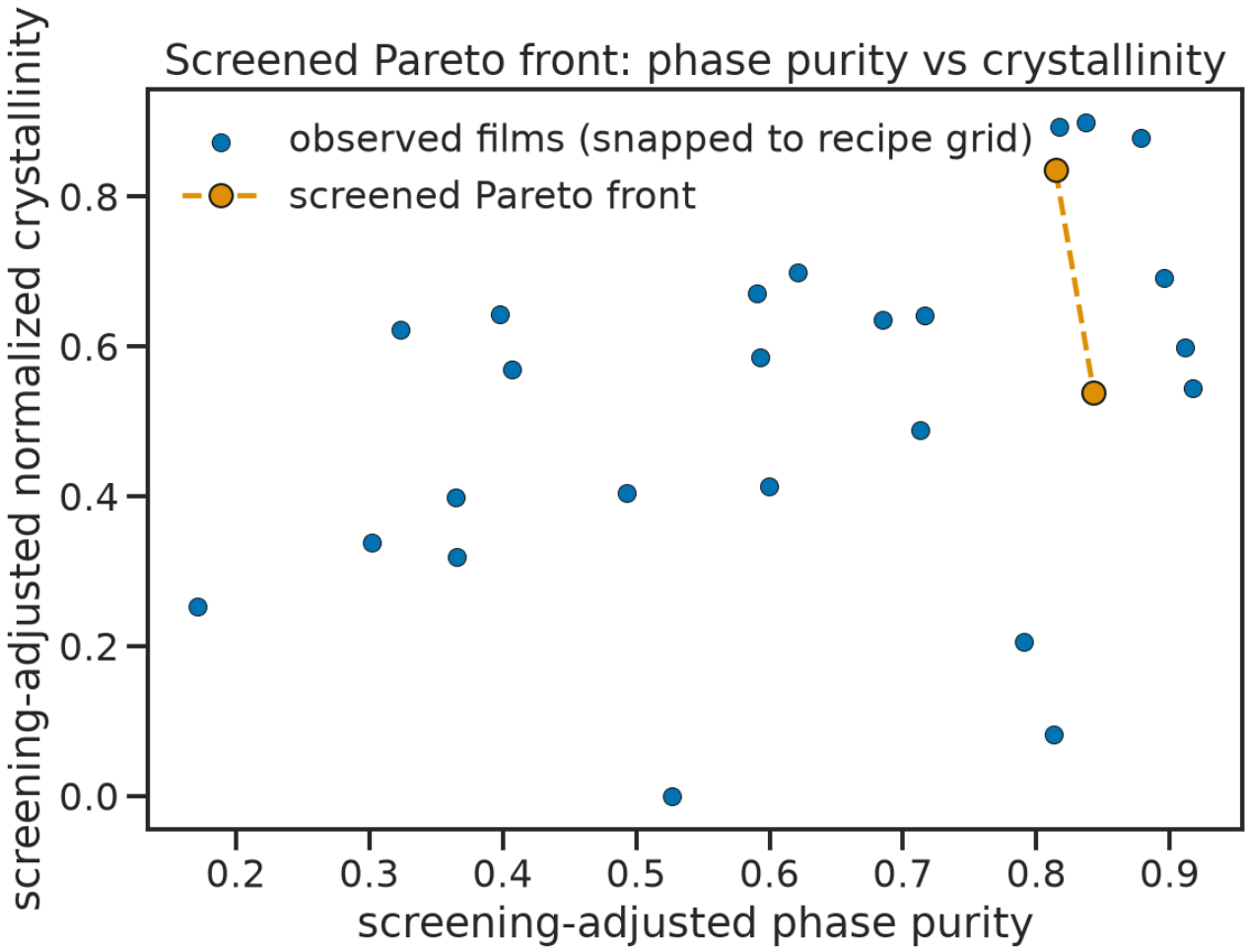
#	Hard bake °C	Selen. temp °C	Selen. time s	CBC ratio	Se duty cycle	Se on ms	Se off ms	Predicted score	GP std
1	250	540	1800	0.680	0.1562	500	2700	0.682	0.170
2	225	550	1800	0.630	0.3125	1000	2200	0.709	0.154
3	250	540	1800	0.660	0.2963	948	2252	0.662	0.179
4	225	540	1800	0.700	0.2931	938	2262	0.663	0.179
5	225	540	1800	0.690	0.2959	947	2253	0.662	0.179

Recommended first experiment: hard bake 250 °C, selenization 540 °C for 1800 s, CBC ratio 0.680, and Se pulse duty cycle 0.1562 (500 ms on / 2700 ms off) (predicted objective 0.68).



5. Two-objective optimization (Ax)

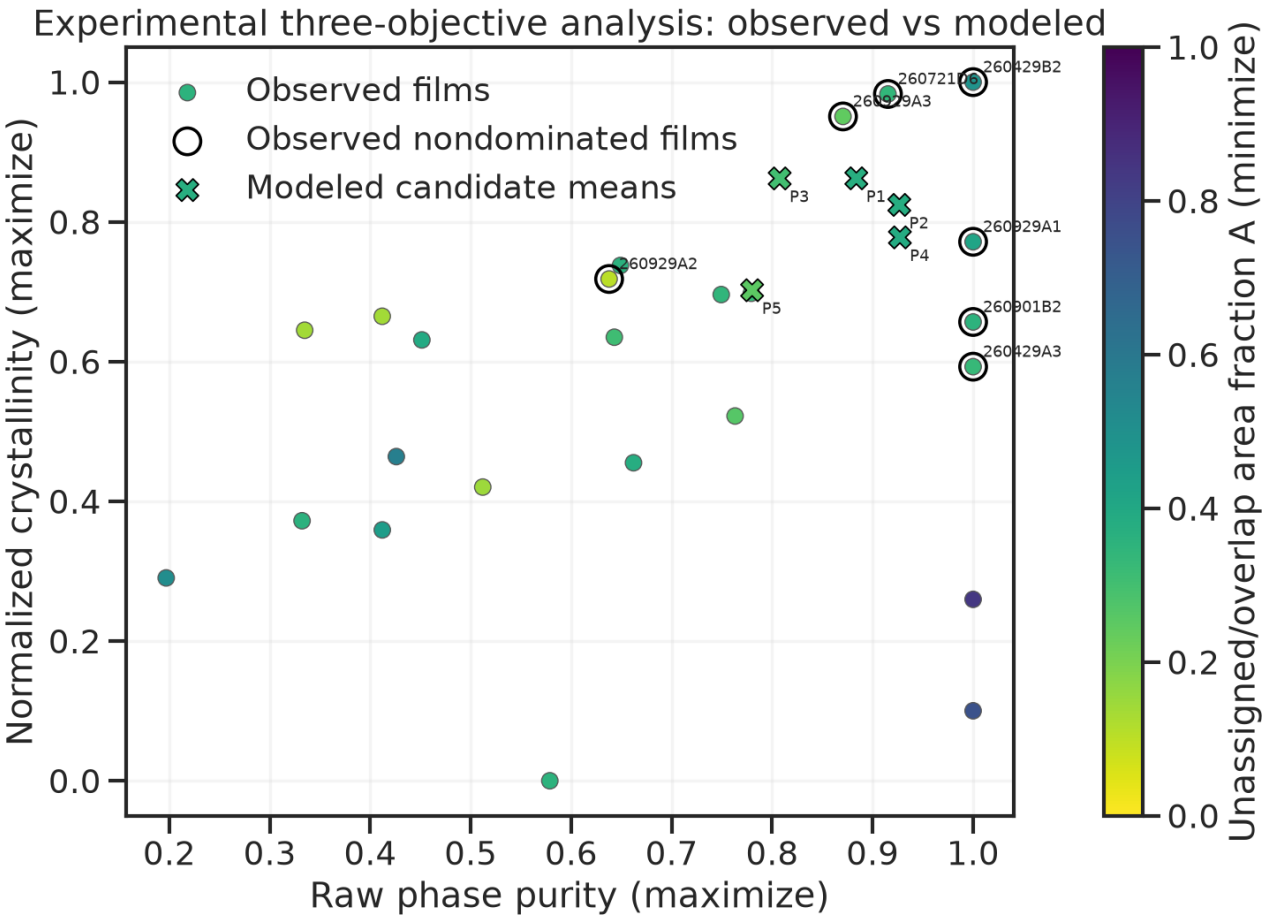
A complementary multi-objective search treats screening-adjusted phase purity and normalized crystallinity as two separate objectives: both are multiplied by the same (1 – 0.25 × A) factor. Ax (BoTorch) fits a Gaussian process per metric and finds the Pareto front via qNEHVI hypervolume maximization. Its five candidate inputs match notebook 3: hard-bake temperature, selenization temperature/time, CBC ratio, and continuous pulse duty cycle. Pulse period is fixed at 3.2 s (0.3125 Hz), so frequency is excluded as a constant. Duty-cycle suggestions are quantized to 1 ms on-time increments; interpolation between the three observed duty levels is exploratory. Numeric process recipes use the same explicit experiment-scale grids, including 25 °C bake steps. This remains a Pareto formulation, not the same scalar objective as notebook 3. A is an exploratory detected-peak-area screening proxy, not a phase fraction or a confirmed assignment of the ~23° feature.



#	Hard bake °C	Selen. temp °C	Selen. time s	CBC ratio	Se duty cycle	Se on ms	Se off ms
1	225	560	1500	0.630	0.3125	1000	2200
2	225	540	1800	0.630	0.1703	545	2655
3	250	560	1800	0.640	0.3125	1000	2200
4	250	530	1500	0.630	0.1897	607	2593
5	250	560	1800	0.630	0.0997	319	2881

6. Experimental three-objective optimization (Ax)

An independent Ax experiment maximizes raw heuristic purity and normalized inverse-FWHM crystallinity while minimizing A directly, without multiplying the first two outcomes by the penalty. Reference thresholds (0, 0, 1.05) define the hypervolume baseline, not feasibility constraints. 7 of 24 complete films are nondominated among the observed films; the displayed model predictions are not measurements and do not establish an exact Pareto surface. Purity and A share peak classifications, and A includes the unassigned ~23° feature; it is not an impurity fraction. This comparison does not replace the production recommendations. In the projection below, the two axes show purity and crystallinity and color represents the third objective A; circles are observed films and crosses are model-predicted candidates.



Observed nondominated films

Observed film	Purity proxy ↑	Crystallinity proxy ↑	A ↓
260429B2	1.000	1.000	0.486
260929A1	1.000	0.771	0.418
260901B2	1.000	0.657	0.356
260429A3	1.000	0.593	0.332
260721D6	0.915	0.983	0.343
260929A3	0.871	0.951	0.244
260929A2	0.638	0.718	0.105

Among these observed films, 260929A2 has the lowest A (0.105) but a purity proxy of 0.638; 260429B2 has the highest normalized crystallinity (1.000) with A 0.486. This illustrates the measured screening-metric trade-off, not a phase-fraction determination.

Three-objective candidate recipes

These candidates are in generation order, not ranked. Predicted mean ± model SEM is evaluated at the quantized recipe and is not an experimental measurement. Pulse period is fixed at 3200 ms; intermediate duty cycles are exploratory.

Proposal	Bake °C	Selen. °C	Selen. s	CBC	Se on / off ms
1	225	550	1800	0.630	513 / 2687
2	250	550	1800	0.630	347 / 2853
3	250	520	1800	0.630	484 / 2716
4	250	550	1800	0.630	705 / 2495
5	225	540	1500	0.630	943 / 2257

Proposal	Purity proxy ↑ mean ± SEM	Crystallinity proxy ↑ mean ± SEM	A ↓ mean ± SEM
1	0.884 ± 0.198	0.862 ± 0.168	0.372 ± 0.175
2	0.927 ± 0.153	0.824 ± 0.190	0.379 ± 0.166
3	0.807 ± 0.147	0.862 ± 0.167	0.301 ± 0.136
4	0.927 ± 0.159	0.778 ± 0.216	0.389 ± 0.172
5	0.780 ± 0.222	0.702 ± 0.221	0.250 ± 0.128